

MView Developer Specification v1.1

Revision focused on the new performance-oriented architecture.

Performance Philosophy

- Never block the UI thread.
- User interaction always has highest priority.
- Cancel obsolete work immediately.
- Prefer immediate visual feedback over maximum image quality.
- Optimize perceived responsiveness rather than raw decoding speed.

Architecture

TMView ■■■■ TJobScheduler ■■■■ TMediaLoader ■■■■ TImageCache ■■■■ TRenderer ■■■■
TNavigator ■■■■ TDirectoryScanner

Responsibilities

TJobScheduler: scheduling, priorities, cancellation, Browse/Skim modes.

TMediaLoader: decode Preview, Screen or Full quality.

TImageCache: immutable decoded images.

TRenderer: paint current image.

TNavigator: navigation state.

TDirectoryScanner: background scanning.

Rendering Pipeline

MediaLoader ↓ ImageCache ↓ Renderer ↓ TMediaView.Paint()

Image Quality Levels

Unknown → Preview → Screen Resolution → Full Resolution

Job Priorities

HIGH: Current image

MEDIUM: Preloading

LOW: Directory scanning

Implementation Roadmap

1. Synchronous viewer
2. ImageCache
3. Background loading
4. JobScheduler
5. Cancellation
6. Directory scanner
7. Preloading
8. Sliding cache

- 9. Browse/Skim modes
- 10. Multiple quality levels
- 11. Predictive preloading
- 12. Progressive refinement